

ZETAZYME COMBI

BIOPOLISHING ENZYME / CELLULASE ENZYME

Product Description:

Zetazyme COMBI is a liquid cellulase formulation designed to enhance fabric texture through abrasion and bio-polishing effects. This intelligent blend of different cellulase enzymes offers versatility, operating efficiently across a broad pH spectrum from 4.0 to 7.0. With an optimal temperature of 52°C to 55°C, Zetazyme COMBI delivers peak performance.

This adaptability allows for its application in both neutral and acidic pH mediums, offering flexibility in fabric treatment processes. Whether aiming for gentle bio-polishing or effective abrasion, Zetazyme COMBI provides the solution for achieving desired fabric finishes with precision and efficiency.

Key Benefits:

- 1. Effective Wide pH Application:** Formulated for application across a wide pH range from 4.0 to 7.0, minimizing the risk of activity loss due to pH fluctuations.
- 2. Dual Action:** Combines abrasion and bio-polishing effects in a single product, eliminating the need for separate inventory for different fabric treatments.
- 3. Acidic and Neutral pH Compatibility:** Can be applied in both acidic and neutral pH conditions, reducing the likelihood of back staining and ensuring consistent results.
- 4. Wide Temperature Activity:** Operates effectively within a broad temperature range of 45°C to 60°C, ensuring sustained activity even with temperature variations.
 - Removes protruding fibres, slubs, hairiness, fluffs, and pills.
 - Prevents material sticking.
 - Softens fabric hand, improves handle, and enhances surface smoothness and structural appearance.
 - Increases flexibility, sew-ability, and resistance to washing-induced pilling.
- 5. Transformational Fabric Improvement:** Converts poor-quality, uneven, napped, or knobby material surfaces into lustrous, soft, elegant fabrics with a fine, high-quality appearance.

Properties:

Chemical Constitution	Formulated with Cellulase enzymes
Ionic Character	—
pH of 1% Solution	Approx. 3.5 to 4.5
Specific Gravity at 20°C	Approx. 1.0 gm/cc
Physical Form	Liquid
Storage Stability	Stable for 6 months at 25°C; strength reduces in contact with air
Stability to Alkalis	Poor

Stability to Acids	Good
Stability to Oxidants	Poor
Stability to Hard Water	Good
Compatibility	Compatible with both non-ionic and anionic products

Applications & Dosage:

Recommended dosage: 0.35% – 0.45% on weight of material. Maintain bath pH at 5.0 and operate at optimum temperature of 52°C – 55°C for best results.

Combined Neutralization, Peroxide Killing & Bio-Polishing

Function	Product	Dosage	Parameters & Conditions
Acid	Acetic Acid	1.0%–1.5%	Auxiliaries dosing – raise temperature to 52°C–55°C and hold for 10–15 minutes
Core-Alkali Neutralizer	Zetacid CAN	0.2%–0.5%	
Peroxide Killer	Zetakill PK	0.2%–0.4%	Add auxiliaries – hold for 10–15 minutes
Bio-polishing Enzyme	ZETAZYME COMBI	0.3%–0.5%	Add auxiliaries – hold for 45–90 minutes

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